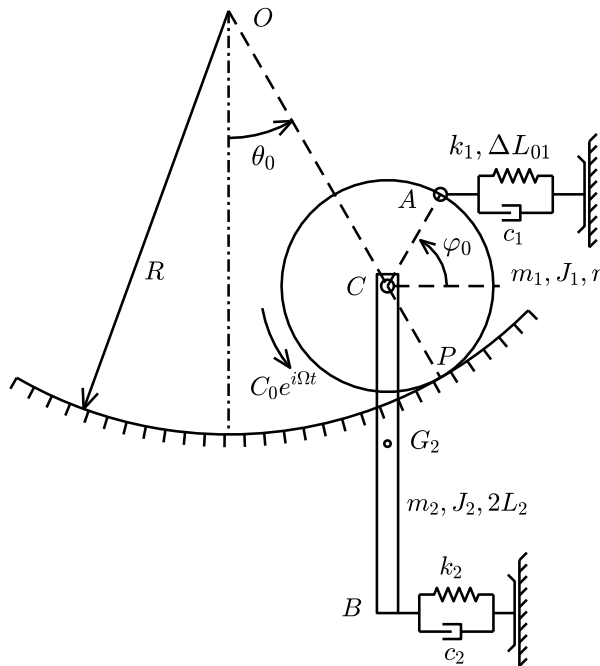


Dynamics of Mechanical Systems

November 08th, 2021



$$\begin{aligned}
 m_1 &= 5 \text{ kg} & c_1 &= 3 \text{ Ns/m} \\
 m_2 &= 5 \text{ kg} & c_2 &= 1 \text{ Ns/m} \\
 J_1 &= 0.15 \text{ kgm}^2 & \theta_0 &= \pi/6 \\
 J_2 &= 0.15 \text{ kgm}^2 & \varphi_0 &= \pi/3 \\
 R &= 1.0 \text{ m} \\
 r &= 0.25 \text{ m} \\
 L_2 &= 0.30 \text{ m} \\
 k_1 &= 5000 \text{ N/m} \\
 k_2 &= 5000 \text{ N/m} \\
 \Delta L_{01} &= 0.0057 \text{ m} \\
 \Delta L_{02} &= 0.0 \text{ m}
 \end{aligned}$$

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding on the circular guide. The rigid bar BC is vertical in the static configuration and is connected to the disk's center through a hinge. The spring and damper system (k_1 , c_1) is connected to point A on the circumference of the disk by a hinge.

The students are requested to:

0. Write on top of each page their surname, name, Person Code and SNxxxx code (see below)
1. Write the equations of motion of the system linearized around the given equilibrium configuration
2. Compute the natural frequencies and modal shapes;
3. Compute the Frequency Response Function (in the range 0 – 15 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the horizontal component of the displacement of the center of mass G_2 and the vertical component of the displacement of point C. Save the graphical results as SNxxxx1.fig and SNxxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxxx are the last four digits of your Person Code (Codice Persona) (e.g. for Bruni Stefano 12345678: BS56781.fig and BS56782.fig)
4. Compute the Frequency Response Function (in the range 0 – 15 Hz, step = 0.01 Hz) applying the harmonic torque $C_0 e^{i\Omega t}$. Output: the tangential and normal components of the contact force in point P. Save the graphical result as SNxxxx3.fig and SNxxxx4.fig.
5. Compute the static preloads ΔL_{01} with $\varphi_0 = \pi/4$.

The answer to points 1, 2 and 5 shall be provided on paper (for point 2 please copy on paper the values of the natural frequencies and eigenmodes produced by the MATLAB script); the answer to points 3 – 4 shall be provided by delivering the MATLAB script and .fig files as a compressed archive uploaded on WeBeeP, see instructions below.

You must upload one single archive file, which should contain your MATLAB script and all the .fig files you want to deliver as part of your solution.

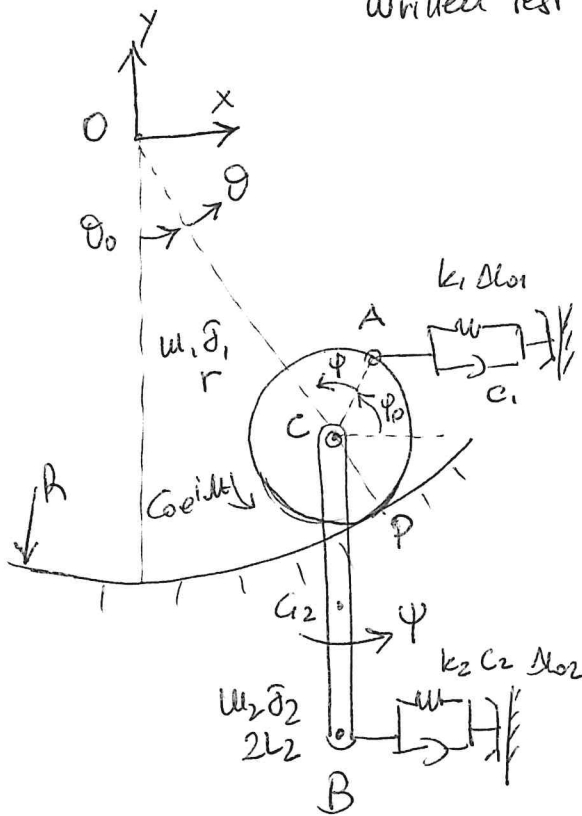
- i.) Accepted formats for the archive files are: .7z .rar .zip;
- ii.) The file must be named SNxxxx with “S” the first letter in your surname, “N” the first letter in your name (in case of multiple surnames/names just consider the first one), “xxxx” the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the name of the file should be: BS5678.ext;
- iii.) The file must be uploaded as an assignment on WeBeeP, the name of the assignment is **Test_8_Nov_2021_B806** or **Test_8_Nov_2021_B807 depending on the room where you are having the test** (make sure you choose the correct folder!);
- iv.) the deadline to upload the assignment will be announced during the written exam.

Please be aware that your answers to the test will not be considered in case:

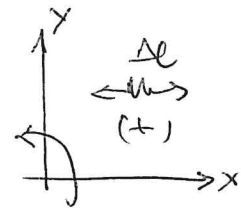
- a. the information stated at point 0 above is missing;
- b. your archive file is uploaded in the wrong assignment folder
- c. the name of the archive file is different from the SNxxxx code as explained above;

DYNAMICS OF MECHANICAL SYSTEMS

Written Test November 08th 2021



• Sign convention:



- 2 rigid bodies = 6 d.o.f. without constr.
- 1 hinge in C: - 2 d.o.f
- rolling without slipping: - 2 d.o.f

$\Rightarrow$ 2 d.o.f. $\underline{x} = \begin{Bmatrix} \theta \\ \psi \end{Bmatrix}$ $\underline{x}_0 = \begin{Bmatrix} \phi \\ \phi \end{Bmatrix}$

$$T = \frac{1}{2} m_1 \dot{\delta l}_1^2 + \frac{1}{2} I_1 \dot{\phi}^2 + \frac{1}{2} m_2 \dot{\delta l}_2^2 + \frac{1}{2} I_2 \dot{\psi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^2 c_i \dot{\delta l}_i^2, \quad V_{el} = \frac{1}{2} \sum_{i=1}^2 k_i \delta l_i^2, \quad V_g = \sum_{i=1}^2 m_i g h_{ci}$$

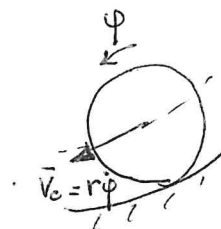
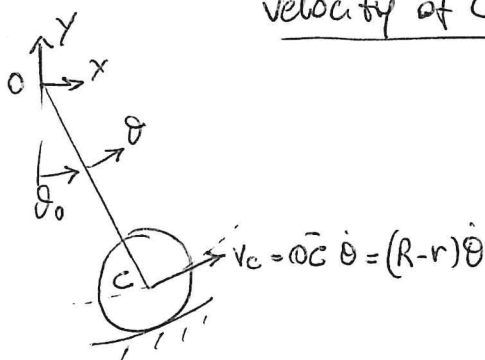
$$\delta W = \vec{C}_{el} \delta \vec{\phi} = C_{el} \delta \phi$$

$$[m_{PH}] = \text{diag}(m_1, I_1, m_2, I_2); \quad \underline{x}_u = \{ \delta l_1, \phi, \delta l_2, \psi \}^T$$

$$[c_{PH}] = \text{diag}(c_1, c_2); \quad [k_{PH}] = \text{diag}(k_1, k_2); \quad \underline{x}_c = \underline{x}_k = \{ \delta l_1, \delta l_2 \}^T$$

$$\underline{F} = \{ C_{el} \delta \phi \}; \quad \underline{x}_a = \{ \phi \}$$

- kinematic relationship between θ and ϕ :
velocity of C:



$$\Rightarrow v_C = -\bar{v}_C; \quad (R-r)\dot{\theta} = -r\dot{\phi}$$

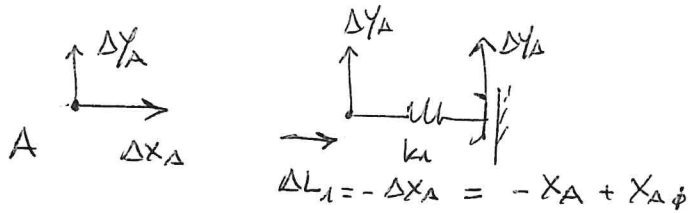
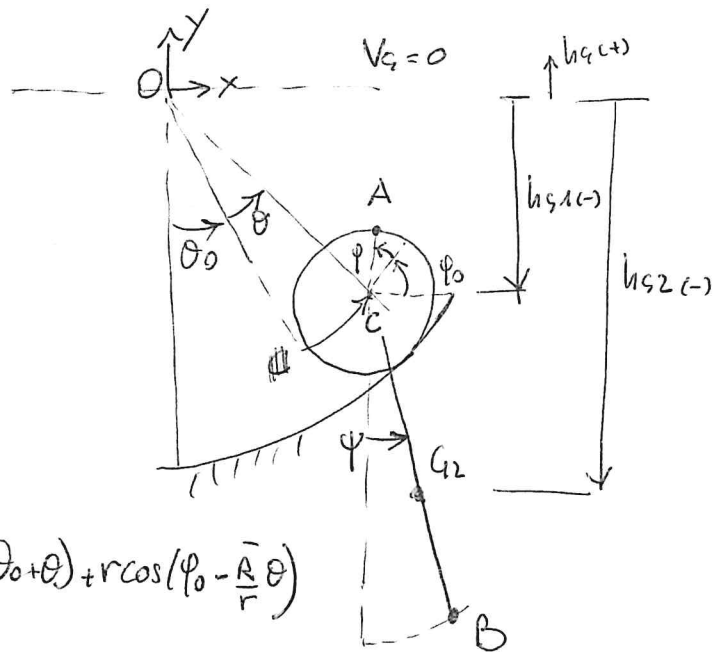
$$\dot{\phi} = -\frac{R-r}{r} \dot{\theta}$$

$$\phi = -\frac{R-r}{r} \theta + \text{const}$$

$$\left\{ \begin{array}{l} \theta=0 \\ \phi=0 \end{array} \right\} \Rightarrow \text{const} = 0 \Rightarrow \boxed{\phi = -\frac{R-r}{r} \theta}$$

Non-linear approach:

$$\begin{cases} \bar{R} = R - r \\ x_c = \bar{R} \sin(\theta_0 + \theta) \\ y_c = -\bar{R} \cos(\theta_0 + \theta) \\ x_{G2} = \bar{R} \sin(\theta_0 + \theta) + L_2 \sin \psi \\ y_{G2} = -\bar{R} \cos(\theta_0 + \theta) - L_2 \cos \psi \\ x_B = \bar{R} \sin(\theta_0 + \theta) + 2L_2 \sin \psi \\ x_A = \bar{R} \sin(\theta_0 + \theta) + r \cos(\varphi_0 + \varphi) = \bar{R} \sin(\theta_0 + \theta) + r \cos(\varphi_0 - \frac{\bar{R}}{r} \theta) \end{cases}$$



Same for k_2 and point B $\rightarrow \Delta L_2 = -\Delta x_B = -x_B + x_{B\phi}$

$$V_C = R\dot{\theta} \rightarrow S_C = R\theta + \text{const} ; \text{const} = 0$$

$$\frac{\partial x_{G2}}{\partial \theta} = \bar{R} \cos(\theta_0 + \theta) ; \frac{\partial x_{G2}}{\partial \psi} = L_2 \cos \psi$$

$$\frac{\partial y_{G2}}{\partial \theta} = \bar{R} \sin(\theta_0 + \theta) ; \frac{\partial y_{G2}}{\partial \psi} = L_2 \sin \psi$$

$$\frac{\partial \Delta L_1}{\partial \theta} = -\bar{R} \cos(\theta_0 + \theta) - \bar{R} \sin(\varphi_0 - \frac{\bar{R}}{r} \theta) ; \frac{\partial \Delta L_1}{\partial \psi} = 0$$

$$\frac{\partial \Delta L_2}{\partial \theta} = -\bar{R} \cos(\theta_0 + \theta) ; \frac{\partial \Delta L_2}{\partial \psi} = -2L_2 \cos \psi$$

$$[A_{u1}] = \begin{bmatrix} \bar{R} & 0 \\ -\bar{R}/r & 0 \\ \bar{R} \cos \theta_0 & L_2 \\ \bar{R} \sin \theta_0 & 0 \\ 0 & 1 \end{bmatrix} \quad [A_{k1}] = [A_C] = \begin{bmatrix} -\bar{R} \cos \theta_0 - \bar{R} \sin \varphi_0 & 0 \\ -R \cos \theta_0 & -2L_2 \end{bmatrix}$$

$$[A_{e1}] = \begin{bmatrix} -\bar{R} & 0 \end{bmatrix}$$

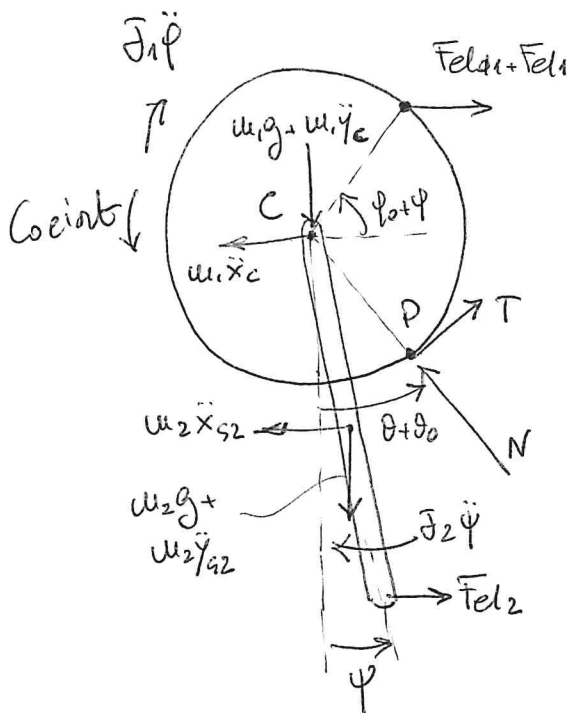
$$h_{s1} = y_c \rightarrow \begin{cases} \frac{\partial^2 h_{s1}}{\partial \theta^2} = \bar{R} \cos(\theta_0 + \theta) \\ \frac{\partial^2 h_{s1}}{\partial \theta \partial \psi} = \frac{\partial^2 h_{s1}}{\partial \psi^2} = 0 \end{cases} \rightarrow [k_{s1}] = m_1 g \begin{bmatrix} \bar{R} \cos \theta_0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$h_{s2} = y_{s2} \rightarrow \begin{cases} \frac{\partial^2 h_{s2}}{\partial \theta^2} = \bar{R} \cos(\theta_0 + \theta) \\ \frac{\partial^2 h_{s2}}{\partial \theta \partial \psi} = 0, \quad \frac{\partial^2 h_{s2}}{\partial \psi^2} = L_2 \cos \psi \end{cases} \rightarrow [k_{s2}] = m_2 g \begin{bmatrix} \bar{R} \cos \theta_0 & 0 \\ 0 & L_2 \end{bmatrix}$$

$$\begin{cases} \frac{\partial^2 \Delta U_1}{\partial \theta^2} = \bar{R} \sin(\theta_0 + \theta) + \frac{\bar{R}^2}{r} \cos(\psi_0 - \theta \frac{\bar{R}}{r}) \\ \frac{\partial^2 \Delta U_1}{\partial \theta \partial \psi} = 0, \quad \frac{\partial^2 \Delta U_1}{\partial \psi^2} = 0 \end{cases} \rightarrow [k_{el II_1}] = k_1 \Delta U_1 \begin{bmatrix} \bar{R} \sin \theta_0 + \frac{\bar{R}^2}{r} \cos \psi_0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Delta U_2 = 0 \rightarrow [k_{el II_2}] = 0$$

• question nr. 4:



$$\sum M_c^{disk} = 0 :$$

$$Tr + Coeint - \bar{F}_{el1} \bar{F}_{el2} - (\bar{F}_{el1} + \bar{F}_{el2}) r \sin(\psi_0 + \psi) = 0$$

$$\sin(\psi_0 + \psi) \approx \sin \psi_0 + \psi \cos \psi_0$$

- $\bar{F}_{el1} \sin \psi_0 \rightarrow$ static Term
- $\bar{F}_{el1} \psi \cos \psi_0 \rightarrow$ linear Term
- $\bar{F}_{el2} \sin \psi_0 \rightarrow$ linear Term
- $\bar{F}_{el2} \psi \cos \psi_0 \rightarrow$ 2nd order Term

$$\downarrow$$

$$T_{tot} = \frac{\bar{F}_{el1} \bar{F}_{el2} - Coeint + \bar{F}_{el1} r \sin \psi_0 + \bar{F}_{el1} r \psi \cos \psi_0}{r}$$

$$\sum F_n = 0 : N - (m_1 g + m_2 g) \cos(\theta_0 + \theta) - (m_1 \ddot{y}_c + m_2 \ddot{y}_{s2}) \cos(\theta_0 + \theta) + (m_1 \ddot{x}_c + m_2 \ddot{x}_{s2}) \cdot \sin(\theta_0 + \theta) - \bar{F}_{el2} \sin(\theta_0 + \theta) - \bar{F}_{el1} \sin(\theta_0 + \theta) - \bar{F}_{el1} \sin(\theta_0 + \theta) = 0$$

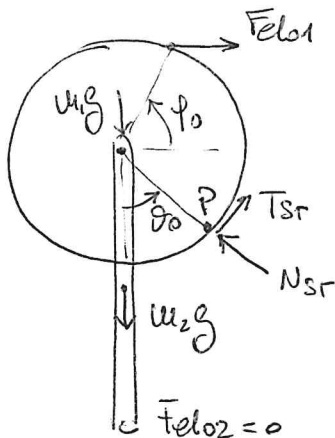
$$\begin{cases} \cos(\theta_0 + \theta) \approx \cos(\theta_0) - \theta \sin \theta_0 \\ \sin(\theta_0 + \theta) \approx \sin(\theta_0) + \theta \cos \theta_0 \end{cases}$$

- $(u_1 g + u_2 g) \cos \theta_0 \rightarrow$ static Term
- $-(u_1 g + u_2 g) \theta \sin \theta_0 \rightarrow$ linear Term
- $(u_1 \ddot{y}_c + u_2 \ddot{y}_{s2}) \cos \theta_0 \rightarrow$ linear Term
- $-(u_1 \ddot{y}_c + u_2 \ddot{y}_{s2}) \theta \sin \theta_0 \rightarrow$ 2nd order Term
- $(u_1 \ddot{x}_c + u_2 \ddot{x}_{s2}) \sin \theta_0 \Rightarrow$ linear Term
- $(u_1 \ddot{x}_c + u_2 \ddot{x}_{s2}) \theta \cos \theta_0 \rightarrow$ 2nd order Term
- $F_{el1} \sin \theta_0 \rightarrow$ static Term
- $F_{el1} \theta \cos \theta_0 \rightarrow$ linear Term
- $(F_{el1} + F_{el2}) \sin \theta_0 \rightarrow$ linear Term
- $(F_{el1} + F_{el2}) \theta \cos \theta_0 \rightarrow$ 2nd order Term

↓

$$N_d q_1 = (u_1 \ddot{y}_c + u_2 \ddot{y}_{s2}) \cos \theta_0 - (u_1 \ddot{x}_c + u_2 \ddot{x}_{s2}) \sin \theta_0 + \\ + (F_{el1} + F_{el2}) \sin \theta_0 - (u_1 g + u_2 g) \theta \sin \theta_0 + F_{el1} \theta \cos \theta_0$$

• question Nr. 5



$$\sum M_P = 0: F_{el1} (r \sin \phi_0 + r \cos \theta_0) + \\ - (u_1 g + u_2 g) r \sin \theta_0 = 0$$

$$F_{el1} = (u_1 g + u_2 g) \frac{\sin \theta_0}{\sin \phi_0 + \cos \theta_0}$$

$$\Delta l_{01} = \frac{F_{el1}}{k_1}$$